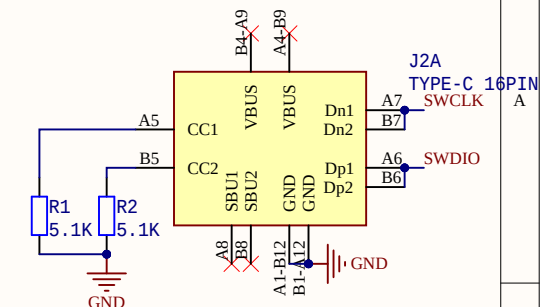
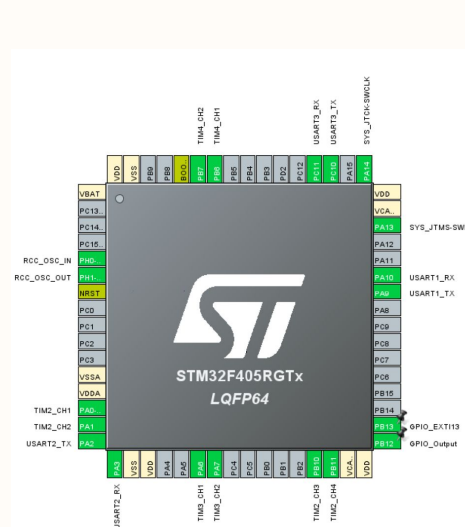
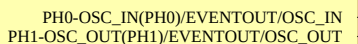
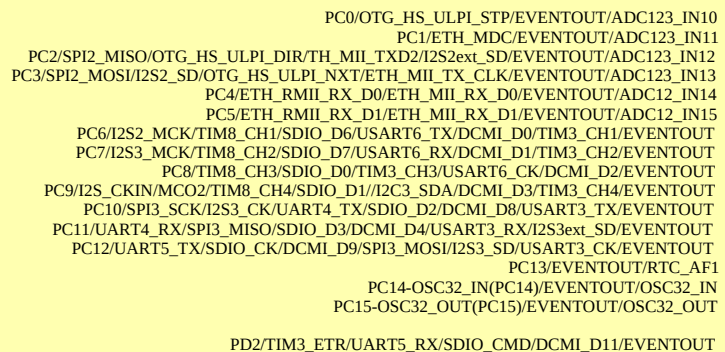
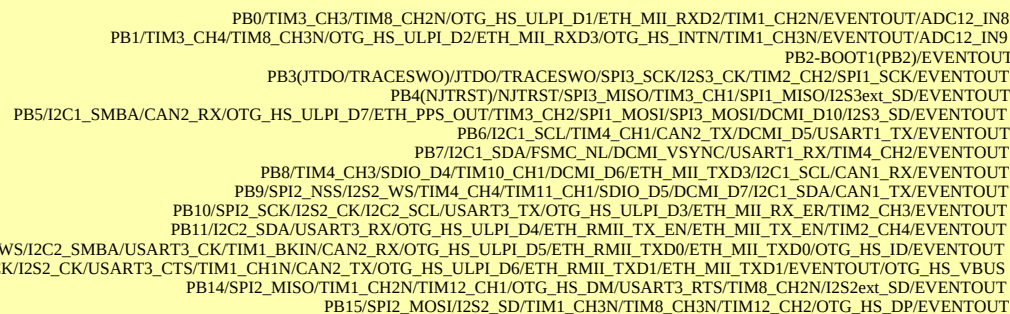
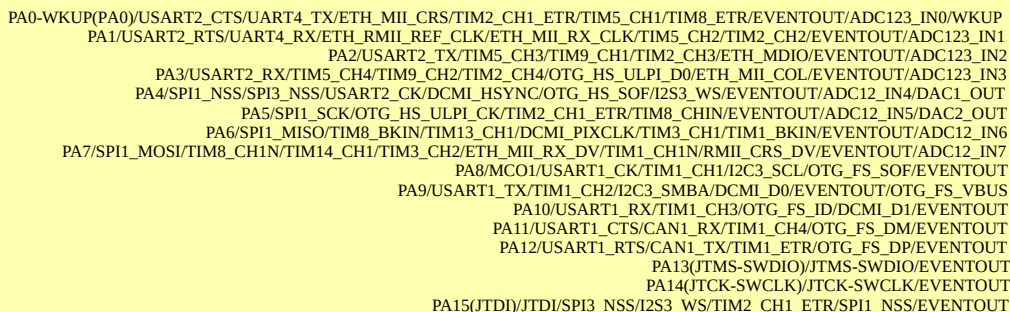
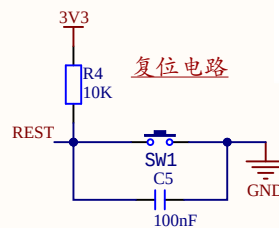
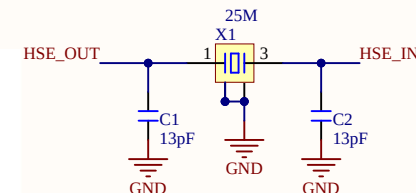


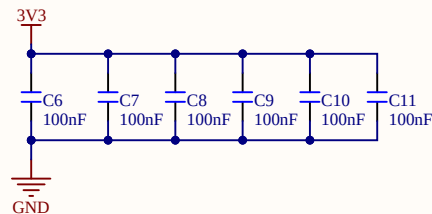
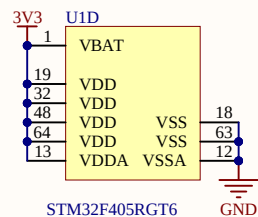
25MHz外部高速晶振



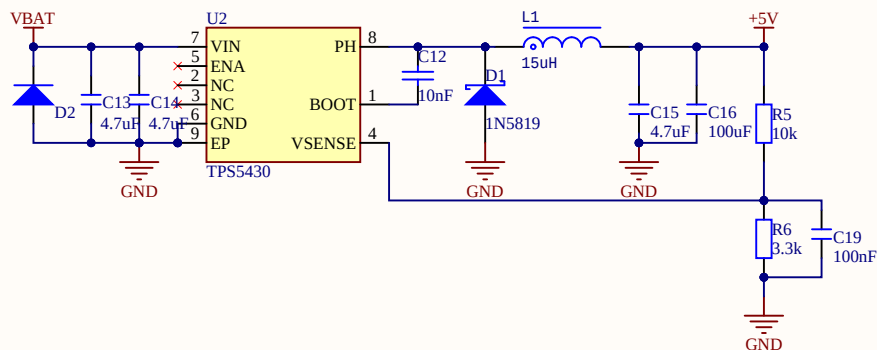
Title 主控芯片		
Size: A4	Number:*	Revision:*
Date: 2024/4/2	Time: 15:27:50	Sheet 1 of 4
File: F:\waj\主控芯片\SchDoc		



去耦电容

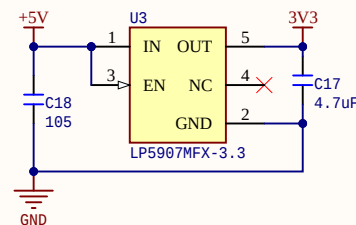


VIN >> 5V

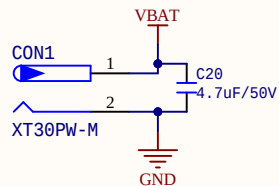


考虑到同时还需要给OpenMV供电，选用输出能力更强的TPS5430，最高可输出5V3A

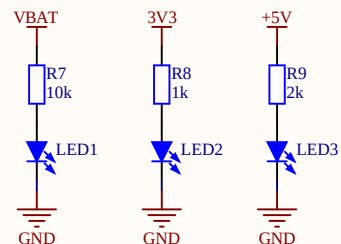
5V >> 3V3



XT30电源接口



电源指示灯



Title 电源

Size: A4

Number:*

Revision:*

Date: 2024/4/2

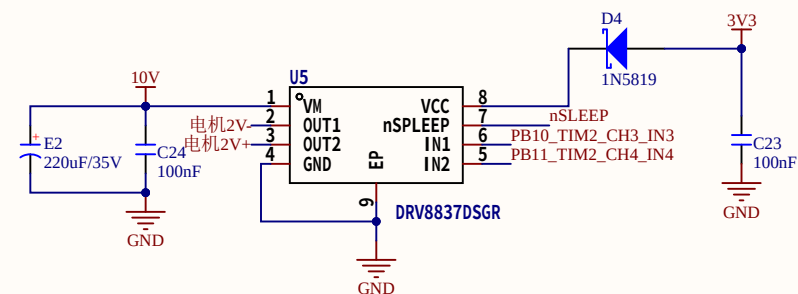
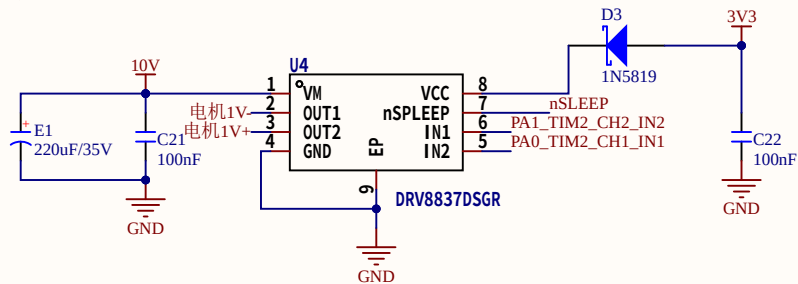
Time: 15:27:51

Sheet 2 of 4

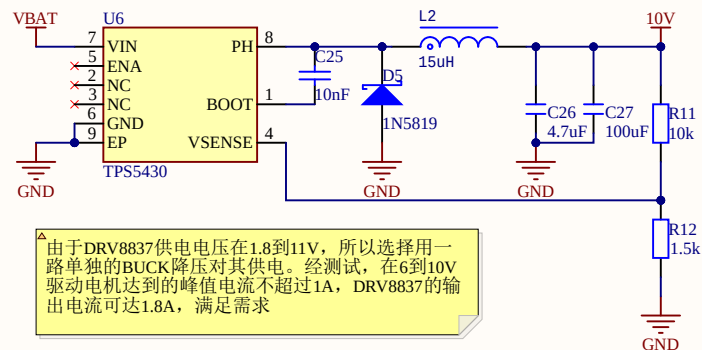
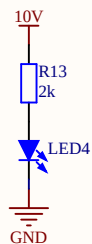
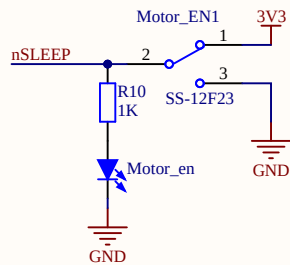
File: F:\waj\电源.SchDoc



H桥驱动电路

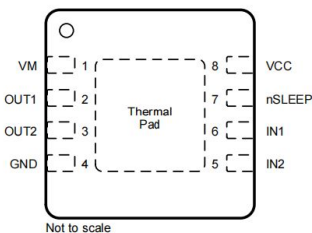


电机驱动使能



由于DRV8837供电电压在1.8到11V，所以选择用一路单独的BUCK降压对其供电。经测试，在6到10V驱动电机达到的峰值电流不超过1A，DRV8837的输出电流可达1.8A，满足需求

DSG Package
8-Pin WSON With Exposed Thermal Pad
Top View



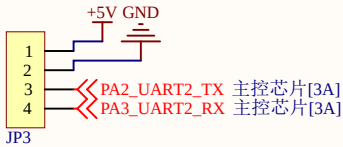
Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
POWER AND GROUND			
GND	4	PWR	Device ground This pin must be connected to the PCB ground.
VCC	8	PWR	Logic power supply Bypass this pin to the GND pin with a 0.1-μF ceramic capacitor rated for VCC.
VM	1	PWR	Motor power supply Bypass this pin to the GND pin with a 0.1-μF ceramic capacitor rated for VM.
CONTROL			
IN1	6	I	IN1 input
IN2	5	I	IN2 input
nSLEEP	7	I	Sleep mode input When this pin is in logic low, the device enters low-power sleep mode. The device operates normally when this pin is logic high. The pin has an internal pulldown resistor to GND.
OUTPUT			
OUT1	2	O	Motor output
OUT2	3	O	Connect this pin to the motor winding.

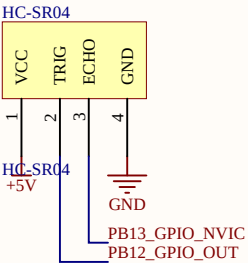
V _{VM}	Motor power-supply voltage	0	11	V
-----------------	----------------------------	---	----	---

Title 电机驱动				
Size: A4	Number:*	Revision:*		
Date: 2024/4/2	Time: 15:27:51	Sheet 3 of 4		
File: F:\waj\电机驱动.SchDoc				

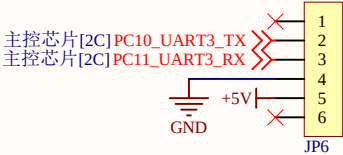
串口 (OpenMV)



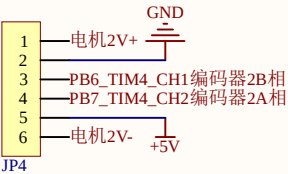
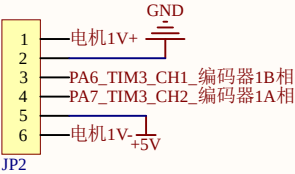
超声波

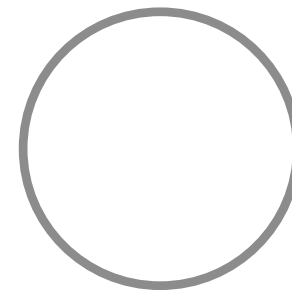
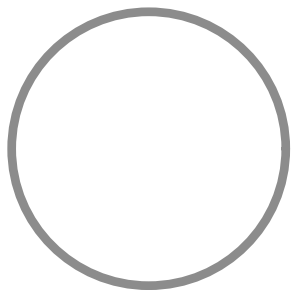


无线通信

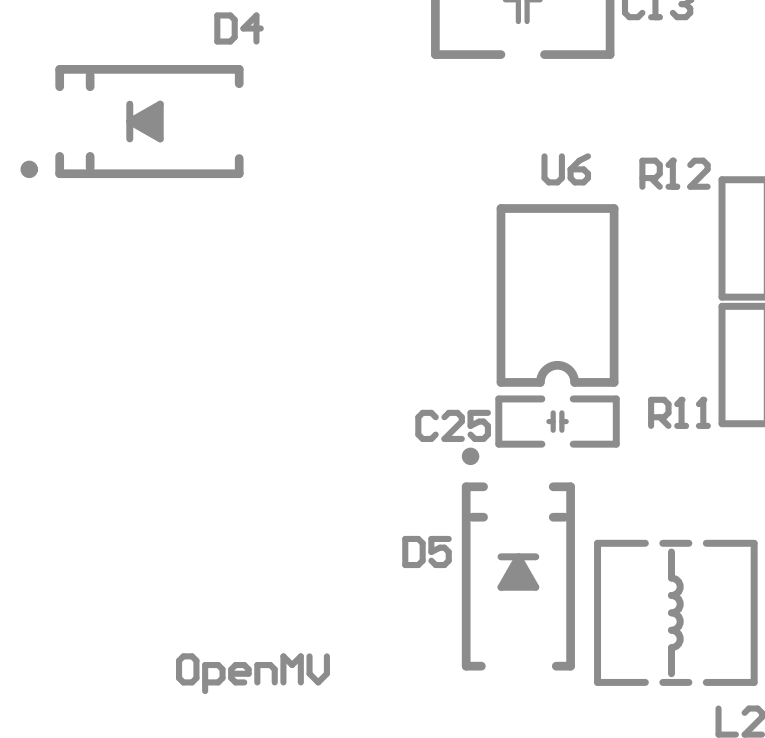
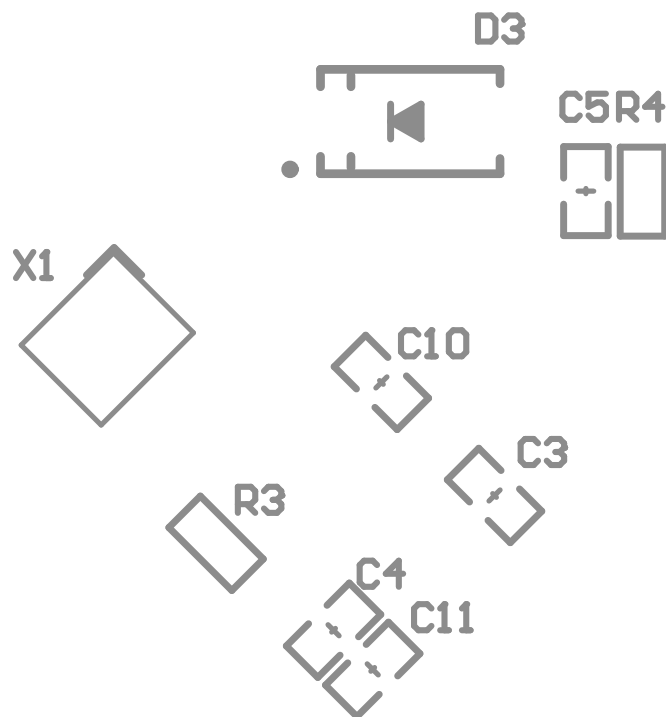


编码器/电机

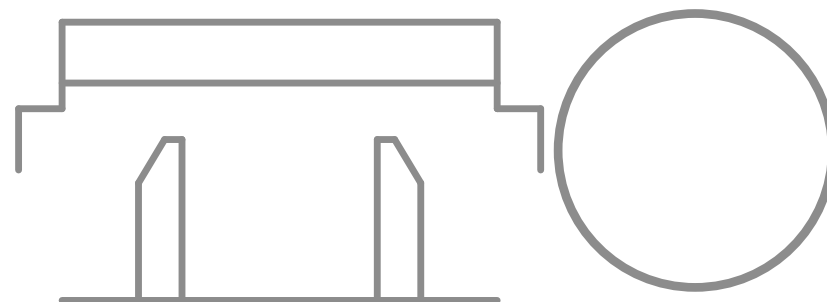
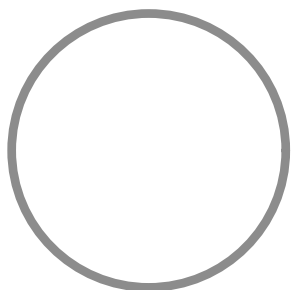


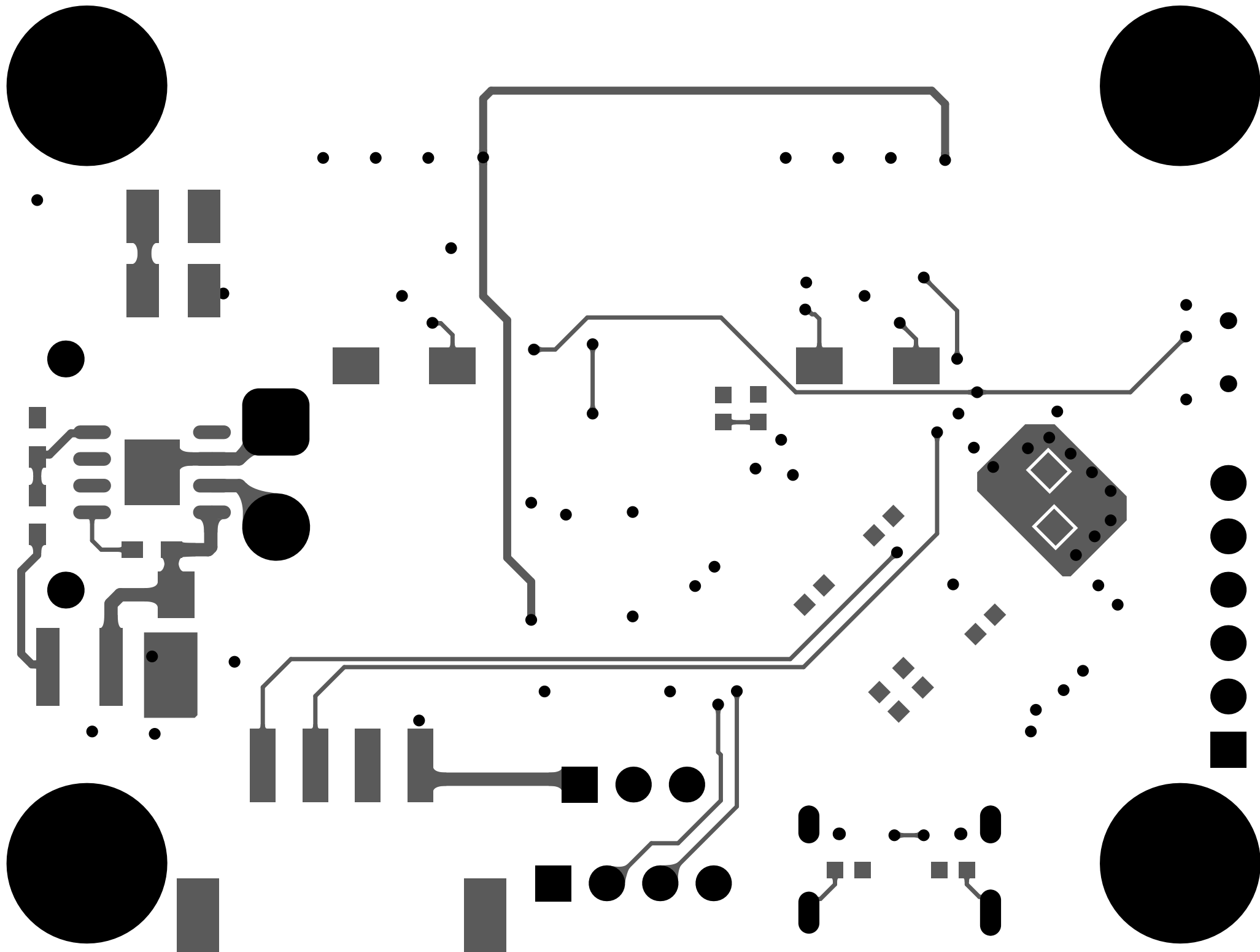


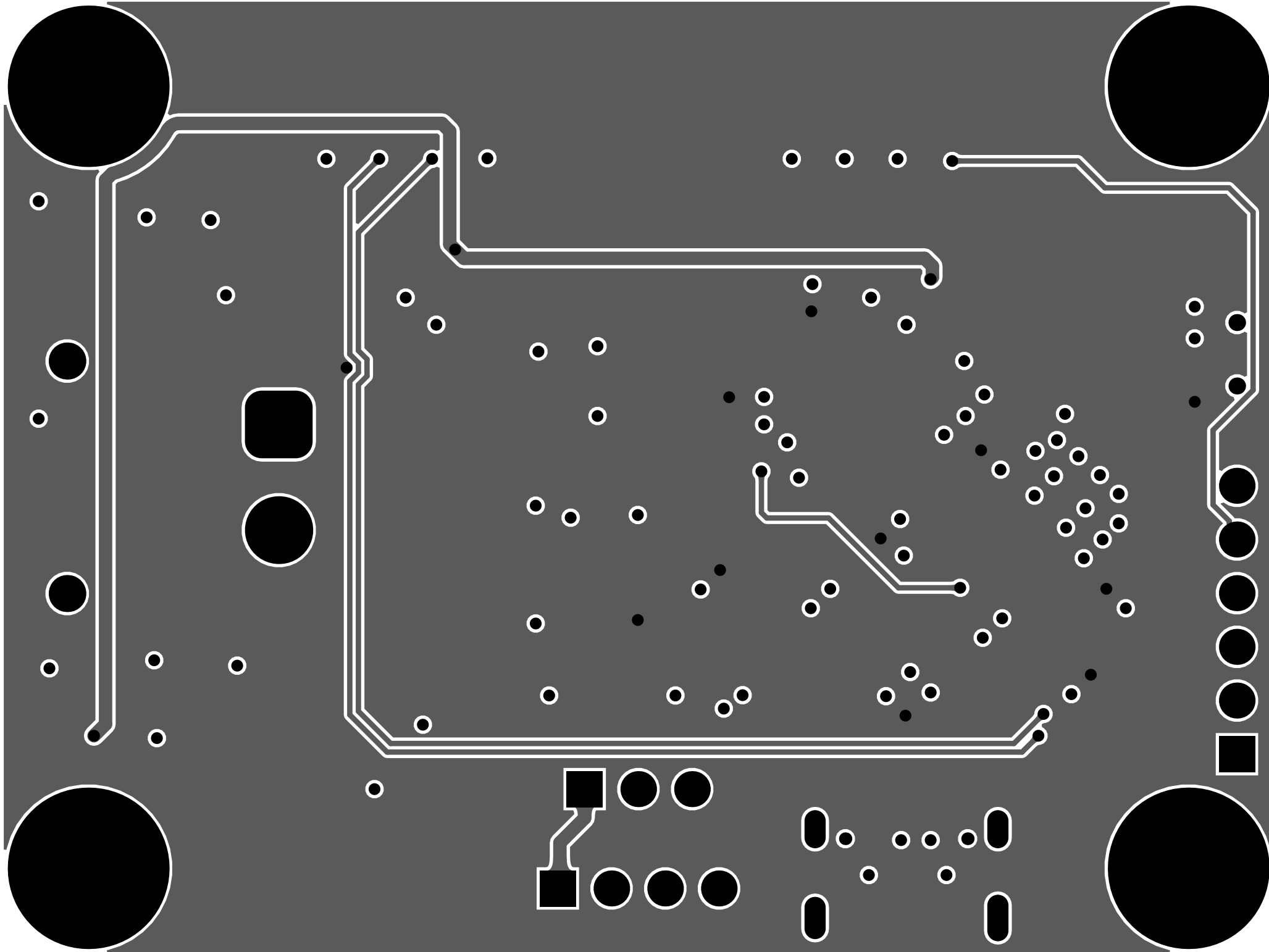
TDPS_Conctrollor

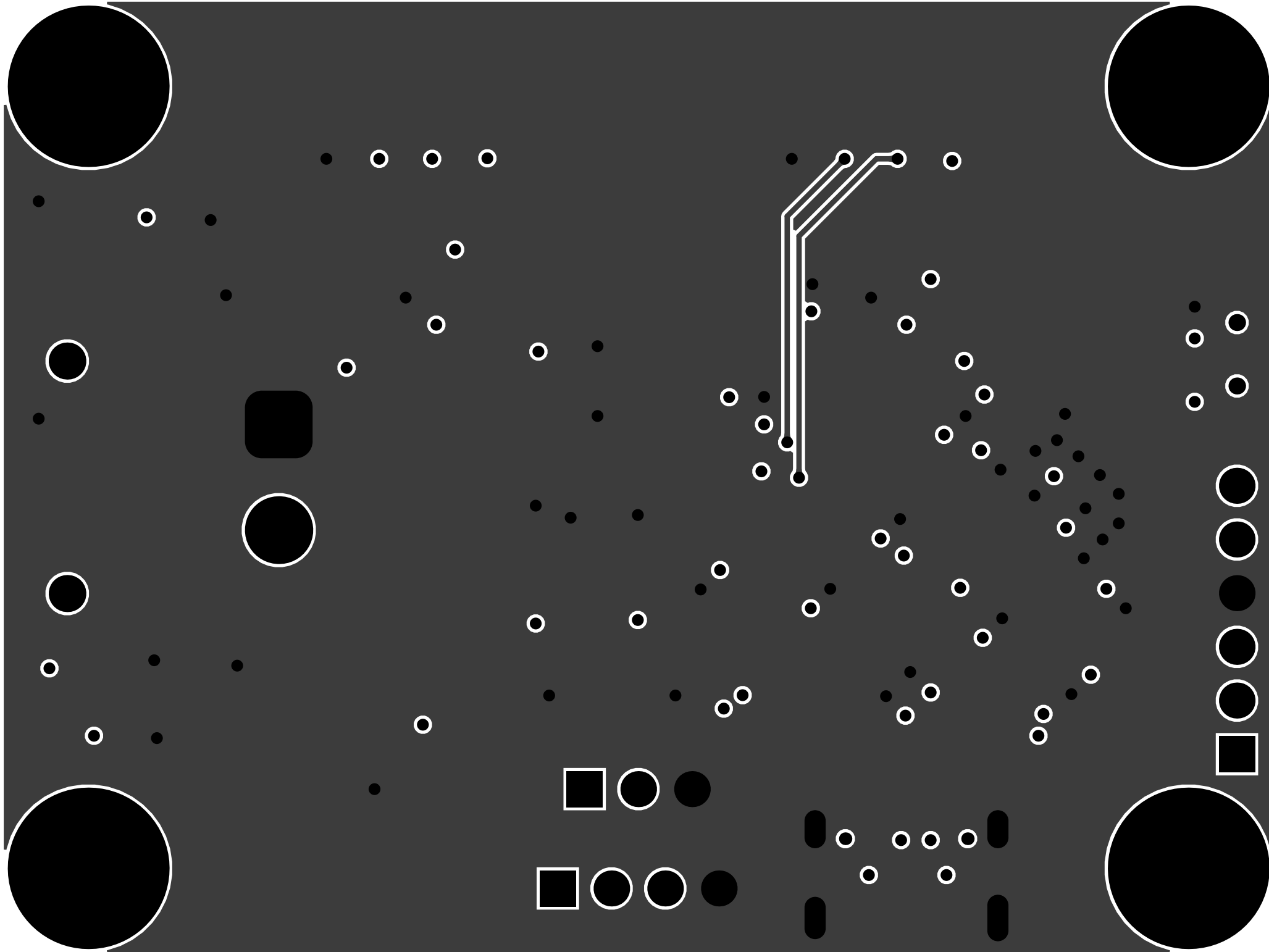


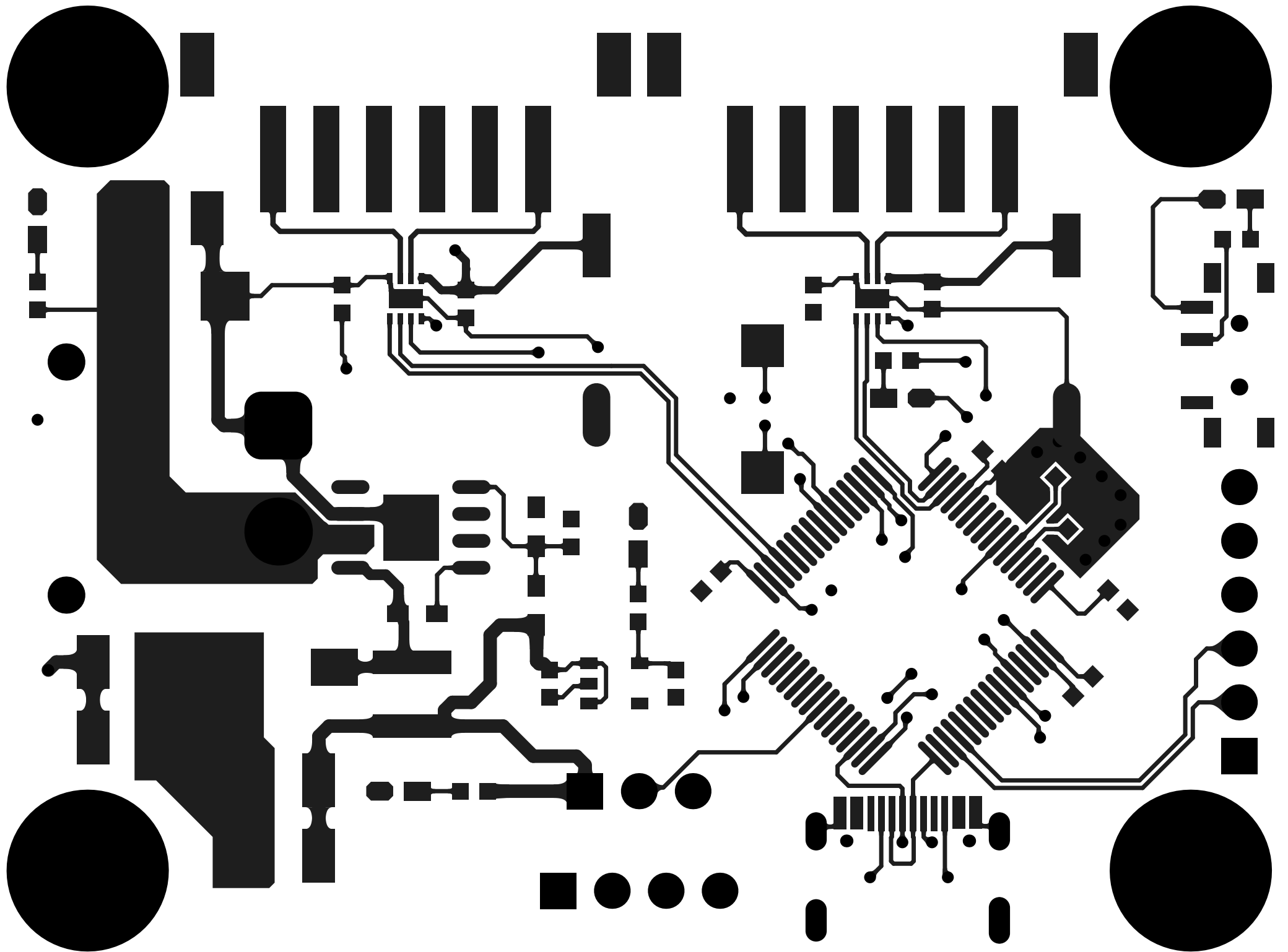
OpenMV

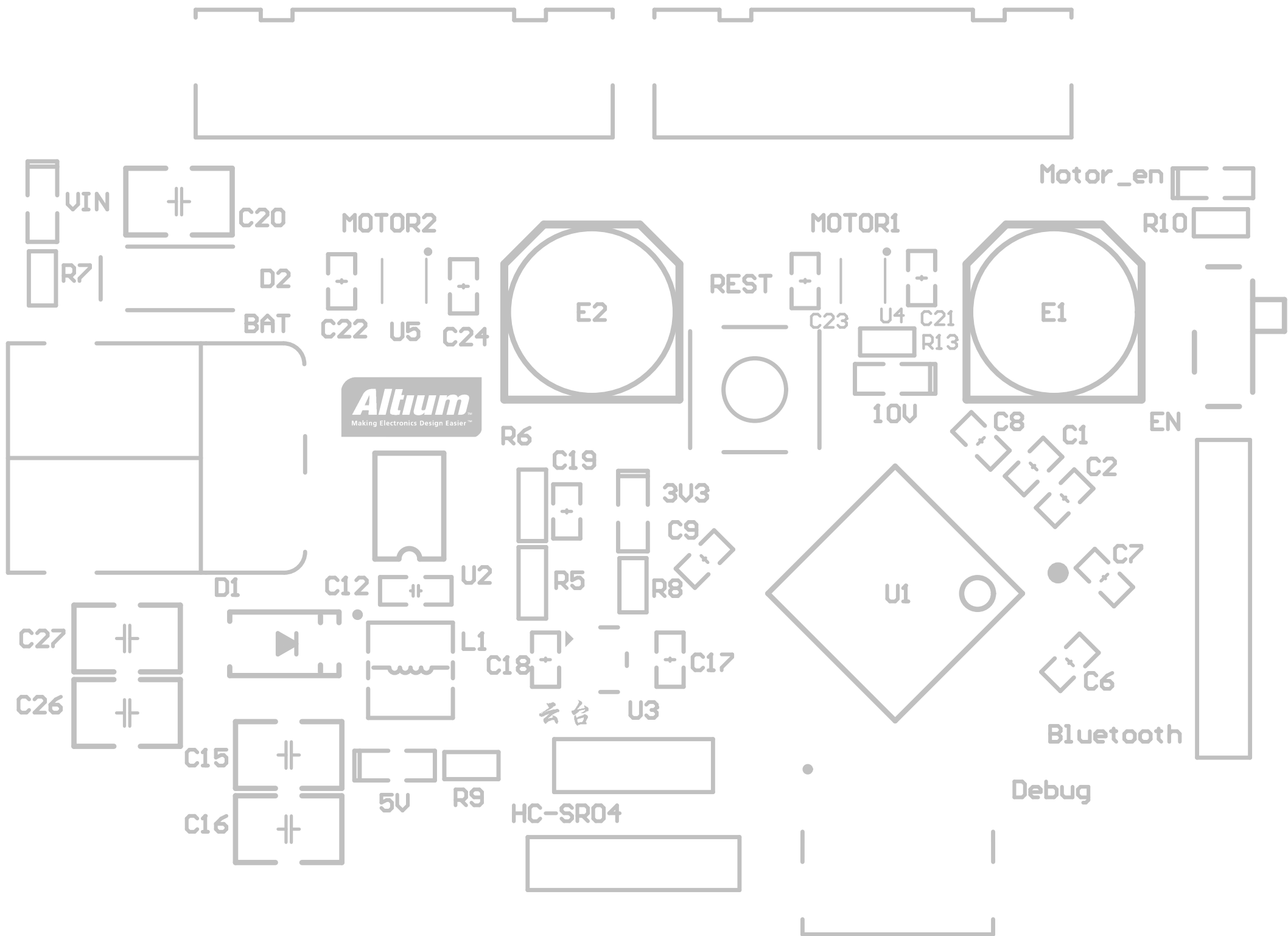












Comment	Description	Designator	Footprint	LibRef	Quantity
13pF	贴片电容	C1, C2	C 0603 L	C	2
2.2u	贴片电容	C3, C4	C 0603 L	C	2
100nF	贴片电容	C5, C6, C7, C8, C9, C10, C11, C19, C21, C22, C23, C24	C 0603_L	C	12
10nF	Capacitor	C12, C25	C 0603 M	Cap	2
4.7uF	贴片电容	C13, C14, C15, C26	C 1210 L	C	4
100uF	贴片电容	C16, C27	C 1210 L	C	2
4.7uF	无极性贴片电容	C17	C 0603 L	C	1
105	无极性贴片电容	C18	C 0603 L	C	1
4.7uF/50V	无极性贴片电容	C20	C 1210 L	C	1
XT30PW-M	XT30PW-M PCB板 专用卧式连接器, 低温升阻燃插头	CON1	XT30PW-M	XT30PW-M	1
1N5819	Schottky Rectifier	D1, D3, D4, D5	SMA	Diode 11DQ03	4
TVS	TVS保护	D2	DIODE-SMA	TVS	1
220uF/35V	贴片电解电容	E1, E2	CM E(8*10.2)	CE_SMD	2
HC-SR04		HC-SR04	HC-SR04	HC-SR04	1
TYPE-C 16PIN	Universal Serial Bus (USB) Shielded I/O Receptacle, Type C, Right-Angle, Surface Mount, Top Mount, Single Row, 16 Circuits, Gold (Au) Plating	J2	Molex_0_MFG	TYPE-C 16PIN 2MD	1
HDR-1X6	6P接插件	JP2, JP4	XH2.54-LS-6P	Header 6	2
HDR-1X4	4P接插件	JP3	XH2.54-WS-4P	Header 4	1
HDR-1X3	3P接插件	JP5	HDR2.54-M-LI-3P	Header 3	1
HDR-1X6	6P接插件	JP6	HDR2.54-M-LI-6P	Header 6	1
15uH	NR磁胶电感	L1, L2	NR4030	L-NR	2
红色	贴片LED	LED1, LED2, LED3, LED4, Motor en	LED 0603W	LED-SMD	5
SS-12F23	1路波动开关	Motor EN1	MSK-12C01-07	SSW-1P	1
5.1K	贴片电阻	R1, R2	R 0603 L	R	2
10k	贴片电阻	R3, R4, R7	R 0603 L	R	3
10k	Resistor	R5, R11	R 0603 M	Res2	2
3.3k	Resistor	R6	R 0603 M	Res2	1
1k	贴片电阻	R8, R10	R 0603 L	R	2
2k	贴片电阻	R9, R13	R 0603 L	R	2
1.5k	Resistor	R12	R 0603 M	Res2	1
6x6x5	2脚贴片6x6无声硅 胶轻触开关	SW1	TS 2PIN 6x6x4.8	TSW SMD-2Pin-6x6	1
STM32F405RGT6	ARM Cortex-M4 32b MCU+FPU, 210DMIP, 1024 kB Flash, 192+4 kB RAM, 51 I/Os, 64- pin LQFP, -40 to +85癆, Tray	U1	LQFP64 10x10_M	STM32F405RGT6	1
TPS5430		U2, U6	SOIC-8-EP	TPS5430	2
LP5907MFX-3.3	250-mA ultra-low- noise low-IQ low- dropout (LDO) linear regulator 5-SOT-23 -40 to 125	U3	Diodes_SOT25_MFG	LP5907MFX-3.3- NOPB	1
DRV8837DSGR	驱动电机类型:有刷 直流电机;电源电 压:1.8V~7V;输出电 流:1.8A;接口:PWM;	U4, U5	DRV8837	DRV8837DSGR	2
25M	4脚无源晶振	X1	OSC 3225-4P	XTAL-4P	1